

IE1005 From Computational Thinking to Programming Part I

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Introduction to IE1005

Computational Thinking and Programming

- Computational Thinking
 - is a thought process
 - involves the **formulation** of a problem
 - then expressing its solution(s) in a way that a computer can effectively carry out
- Programming
 - The computer needs **detailed and exact instructions** to carry out the steps needed to solve a problem.
 - These instructions must be **Programmed** or coded (i.e. written) using a **programming language** (C, Java, Python) and they form a computer program.

Intended Learning Outcomes

Explain the **programming concepts** taught in this course.



Write **basic computer programs** based on the programming language taught in the course.



Apply the concepts learnt on case studies/problem-based scenarios:
translating **Computational Thinking to actual Computer Programs**

Reference Books

- Textbook
 - Harry H. Cheng, *C for Engineers and Scientists : An Interpretive Approach*, McGraw Hill, 2010
- Other References
 - Deitel, Paul J. and Deitel, Harvey M., *C : How to Program* , 9th Edition, Pearson Prentice Hall, 2022
 - Etter, Delores M., *Engineering Problem Solving with C*, 4th Edition, Pearson Prentice Hall, 2013
 - Dey, Pradip. and Ghosh, Manas., *Computer Fundamentals and Programming in C*, 2nd Edition, Oxford University Press, 2013

Outline

Part I by
Prof. Lee Seok Woo

Computers

Overview of C

Fundamental Data Types

C Operators

Standard Library Functions

Decision Making

Loops

Part II by
Prof. CHANG Wonkeun

Functions

Modular Programming

Arrays

Variable scope

Pointer

Lecture: 2 hours per week

Week 1 to Week 12

Tutorial: 2 hours per week

Week 2 to Week 13

Part 1 Schedule

Week	Attend Physical Lecture	Attend Tutorial class
1	Introduction to IE1005 L01: Computers & Programming	No Tut/Practical
2	L02: Overview of C	Tut/Prac 1: Introduction to Computers & Programming
3	L03: Fundamental Data Types	Tut/Prac 2: Overview of C
4	L04: C Operators L05: Standard Library Functions	Tut/Prac 3: Fundamental Data Types
5	L06: Decision Making	Tut/Prac 4: C Operators, Standard Library Functions Home Assignment #1
6	L07: Loops	Tut/Prac 5: Decision Making
7	Decision making and Loops	Tut/Prac 6: Loops Practical Test #1

Part 2 Schedule

Week	Attend Physical Lecture	Attend Tutorial class
8	Functions	Tut/Prac 7: Decision making and Loops
9	Modular Programming	Tut/Prac 8: Functions Home Assignment #2
10	Arrays	Tut/Prac 9: Modular Programming
11	Variable scope	Tut/Prac 10: Arrays 1 Practical Test #1
12	Pointers	Tut/Prac 11: Arrays 2
13	Final Quiz	Tut/Prac 12: Variables scope & Pointers (Online tutorial)

Assessment Summary

• Class Participation	: 10%	
• week 5, Home Assignment 1	: 10%	} Marks will be released in NTULearn.
• week 7, Practical Test 1	: 20%	
• week 9, Home Assignment 2	: 10%	
• week 11, Practical Test 2	: 20%	
• <u>week 13, Final Quiz</u>	<u>: 30%</u>	
• Total	: 100%	

Assessment Components (1)

Class Participation (10%)

- To check your understanding, you have to answer tutorial questions in class
- You have to work on practical questions at home and when time allows in class
- Professional attitude. (e.g. no playing game)

2 Home Assignments (10% + 10%): week 5 and week 9

- You will be given a question to make a code.
 - Submit code and video recording of your explanation of the code.
- You will be given 2 such exercises throughout the semester to attempt
- Copying from others is **DISALLOWED** and will incur **HEAVY** penalty

Assessment Components (2)

2 Practical Tests (20% + 20%)

- Restricted Open-book test
- You have to complete and submit 1 Programming Question per Test in class
- You will have to attempt 2 such tests, in Week 7 and in Week 11
- More details will be given in Tutorial & Practical Lessons by respective Tutors

Quiz (30%)

- This is a common quiz/test that you will all have to attempt:
 - **Book lecture time in week 13.**
- It is Closed-book test. More details will be announced in due time

Resources

- More Practice questions and exercises are placed at the NTULearn IE1005 Course site. Please attempt them.

The screenshot shows the NTULearn interface for the course IE1005-FR COMPUTATIONAL THINKING TO PRO (LEC). The left sidebar contains a menu with 'Content' circled in red and 'Course Media' circled in blue. The main content area lists several folders: 'Lecture and Tutorial Material Part 1', 'Lecture and Tutorial Material Part II', 'Handson', 'Week 1: Common Pre Topic 1 and 2', and 'Week 13: Common Post Topic 1 and 2'. Annotations include: a red arrow pointing from the red circle to the 'Content' header; an orange box containing a checklist for 'EXTRA Questions and Exercises Part 1' (C Operators / Standard Library Functions, Decision Making, Loops) with an arrow pointing to 'Lecture and Tutorial Material Part 1'; a yellow box containing a checklist for 'Extra Exercises Part 2' (Function 1, Modular Programming) with an arrow pointing to 'Handson'; and a blue box containing the text 'The recorded videos are available at Course Media.' with an arrow pointing from the blue circle to the 'Course Media' link in the sidebar.

IE1005-FR COMPUTATIONAL THINKING TO PRO (LEC)

Announcements

Content

Discussion Board

Course Media

Groups

Tools

Content

Lecture and Tutorial Material Part 1

Lecture and Tutorial Material Part II

Handson

Week 1: Common Pre Topic 1 and 2

This folder consists of lecture notes and learning activities modules:

- **Course Introduction:** Computational Thinking
- **Programming Languages:** Machine-language

Week 13: Common Post Topic 1 and 2

This folder consists of lecture notes and learning activities modules:

- **Computing Trends**
- **Computer Ethics**

- ☐ EXTRA Questions and Exercises Part 1
 - ☐ C Operators / Standard Library Functions
 - ☐ Decision Making
 - ☐ Loops
- ☐ Extra Exercises Part 2 are:
 - ☐ Function 1
 - ☐ Modular Programming
- ☐ The recorded videos are available at Course Media.

An Advice

- You are going to learn C programming language.
- Just like learning any new language like Japanese, French etc.
- **PRACTISE and USE C programming language regularly!**
- So,
 - i) Attempt all Tutorial / Practical exercises before the lesson
 - ii) Clarify your doubts with your course-mates, friends, tutors or lecturers
EARLY

Special Announcement

- Stick to your registered Tutorial Group
- **Do not** attend another class.
 - Limited number of computers
- You can only take the Continual Assessments in your registered class.
- Please check your NTU e-mailbox regularly as your tutor/lecturers will communicate with you via your NTU email account.

Key Contact Personnel

- Course matters: Course Coordinator
 - Email: sw.lee@ntu.edu.sg
 - or MS Teams
- Part 1 Lecture: **Prof. LEE Seok Woo**
- Part 2 Lecture: **Prof. CHANG Wonkeun**
 - Email: wonkeun.chang@ntu.edu.sg
 - or MS Teams
- Tutorial lessons, CAs: **your tutor and Teaching Assistants (TAs)**